

Credit Card Fraud Detection

XGBoost, Chunk Evaluation & Drift Analysis

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- 6 Experiments and Results
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Figure: Fraud rate by time-sorted chunk

Problem Statement

Challenge: Fraud detection is an imbalanced classification problem where fraudulent transactions are rare but high-impact.

Key issues:

- Accuracy is misleading because the non-fraud class dominates.
- Fraud patterns can change over time, causing concept or distribution drift.
- A fixed threshold can become invalid even when model ranking remains good.
- False negatives lose fraud cases; false positives create unnecessary alerts.

Goal: Build an XGBoost-based fraud detector and evaluate whether low F1 scores are caused by concept drift or threshold/calibration drift.

Dataset Overview

Dataset: Time-sorted credit card transaction data split into 16 parquet chunks.

Attribute	Value
Total transactions	8,580,255
Fraud transactions	94,806
Overall fraud rate	1.105%
Number of chunks	16
Rows per chunk	$\approx$ 536K
Evaluated chunks	02–16

Note: Chunk 16 has no fraud cases in the test segment, so it is excluded from model-quality conclusions.

Fraud Rate Over Time



Figure: Fraud rate varies across time-sorted chunks, indicating possible prior or distribution shift.

Chunk 01: Data Quality Overview

Source used in notebook:

data_chunks_16/transactions_time_sorted_part_01.parquet

Attribute	Value
Rows	536,266
Columns	44
Missing values	0 across all columns
Unique transactions	536,266
Unique cards/accounts	8,757
Categories	14
Merchants	655
Jobs	639
Transaction period	2022-05-06 23:00:01 to 2022-05-30 02:24:20

EDA implication: data is clean enough for modeling, but high-cardinality categorical variables require encoding.

Chunk 01: Target Imbalance



Class	Count	Rate
Non-fraud	530,583	98.9403%
Fraud	5,683	1.0597%

- Accuracy is not a meaningful metric.
- PR-AUC, recall, precision, and F1 are more useful.
- XGBoost needs imbalance handling through class weight or undersampling.

Chunk 01: Datetime and Age

Variable	Minimum	Maximum
Date of birth	1927-07-06	2008-07-03
Transaction datetime	2022-05-06 23:00:01	2022-05-30 02:24:20
Age	14	95

Age statistic	Mean	Std	Q1	Median	Q3
Age	44.23	17.90	30	41	55

Feature decision: use ordinal `age_bin_ord` instead of raw age to capture non-linear age-group effects.

Time EDA: Hour of Day

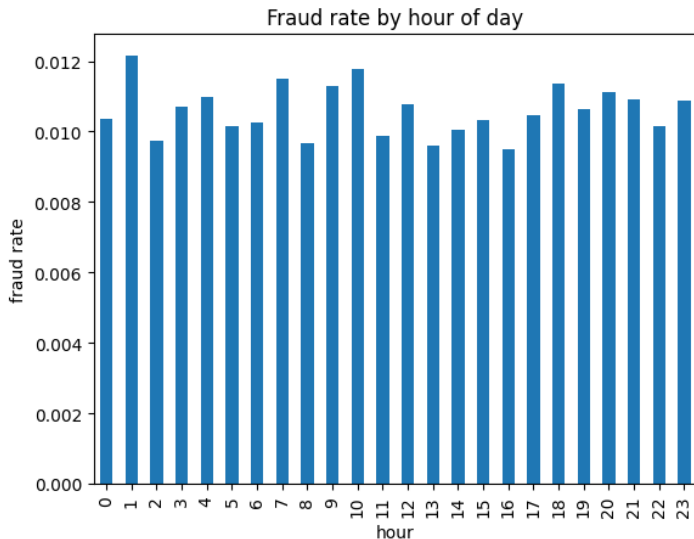


Figure: Fraud rate by transaction hour in chunk 01

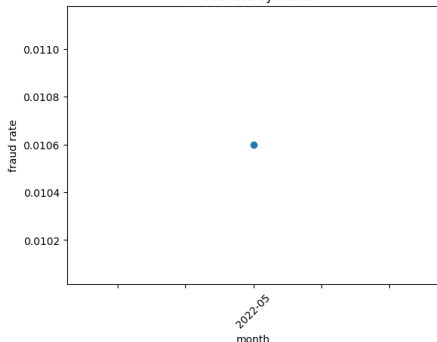
Time EDA: Weekday vs Weekend

Segment	Total	Fraud	Fraud rate
Weekday	285,937	3,778	1.3213%
Weekend	250,329	1,905	0.7610%

- Fraud appears more often on weekdays in chunk 01.
- This contradicts a simple rule such as “fraud happens mostly at night/weekend”.
- `trans_date_is_weekend` is still useful, but the direction should be learned by the model.

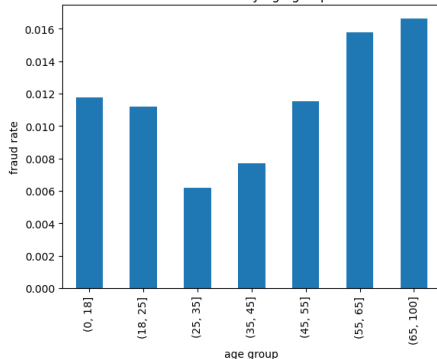
Time EDA: Month and Age Group

Fraud rate by month



Month-level fraud rate

Fraud rate by age group



Fraud rate by age bin

Older age groups show higher fraud rates in this chunk: age 55–65 has 1.58%, age 65–100 has 1.66%.

Categorical EDA: Cardinality

Column	Type	Unique	Null count	Null rate
merchant	string	655	0	0.0%
job	string	639	0	0.0%
category	string	14	0	0.0%
gender	string	2	0	0.0%

Encoding decision:

- category: one-hot encoding because cardinality is low.
- merchant, job: frequency encoding because cardinality is high.
- gender: binary encoding.

Categorical EDA: Category and Gender

Category	Total	Fraud	Rate
grocery_pos	1,394	1,394	100.0%
shopping_net	1,299	1,299	100.0%
misc_net	744	744	100.0%
shopping_pos	659	659	100.0%
gas_transport	458,338	465	0.1015%
grocery_net	72,804	94	0.1291%

Gender	Total	Fraud	Rate
M	266,428	2,963	1.112%
F	269,838	2,720	1.008%

Warning: several categories are almost deterministic in chunk 01, so metrics must be checked across future chunks to avoid over-trusting category artifacts.

Categorical EDA: Merchant and Job

Merchant signal

- Top fraud merchant rates are only around 0.2–0.3%.
- Merchant alone is weaker than category.
- Frequency encoding is kept as a compact signal.

Top job fraud rates

Job	Fraud rate
Scientist, product/process dev.	9.82%
Systems developer	7.00%
Theatre manager	6.38%
Copywriter, advertising	5.99%
Holiday representative	5.42%

Insight: job has stronger discriminative signal than merchant in chunk 01.

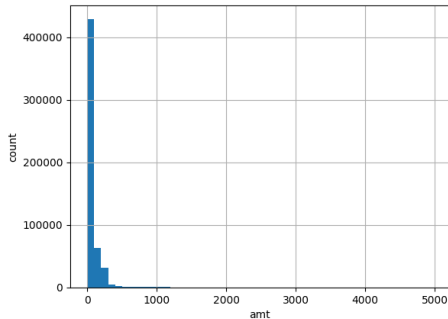
Numeric EDA: Fraud vs Non-Fraud

Feature	Non-fraud mean	Fraud mean	Non-fraud median	Fraud median
amt	78.57	534.61	61.51	
trans_time_hrs	11.50	11.48	12.00	
trans_date_is_weekend	0.468	0.335	0.00	
merchant_risk_30_day	16.12	16.25	17.00	
customer_num_trans_30_day	13.00	13.04	13.00	

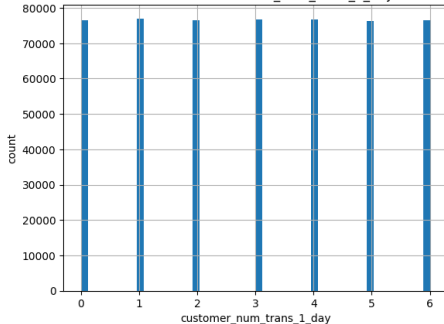
- Amount is the strongest numeric signal: fraud mean is about 6.8x non-fraud mean.
- Rolling customer/merchant counts have much weaker univariate separation.

Numeric Distributions I

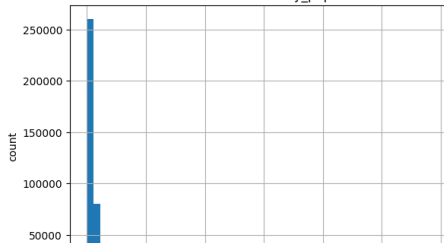
Distribution of amt



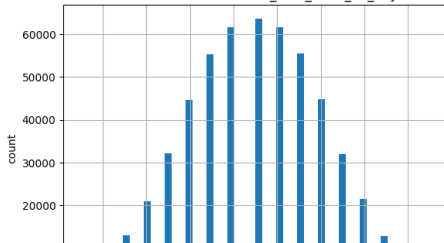
Distribution of customer_num_trans_1_day



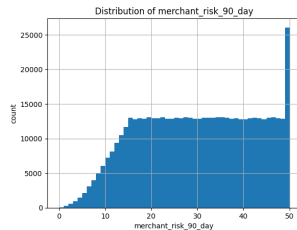
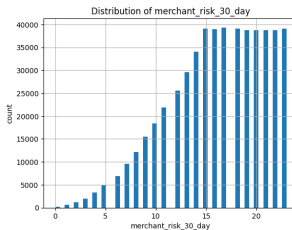
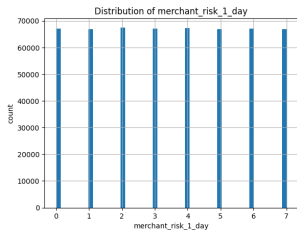
Distribution of city_pop



Distribution of customer_num_trans_30_day

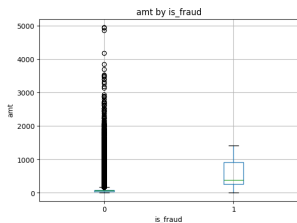


Numeric Distributions II

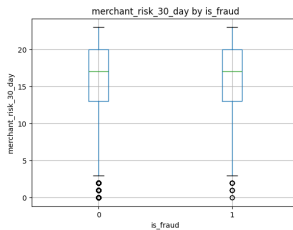


Merchant risk features are retained for interactions, although their univariate separation is weak.

Boxplots by Target



amt

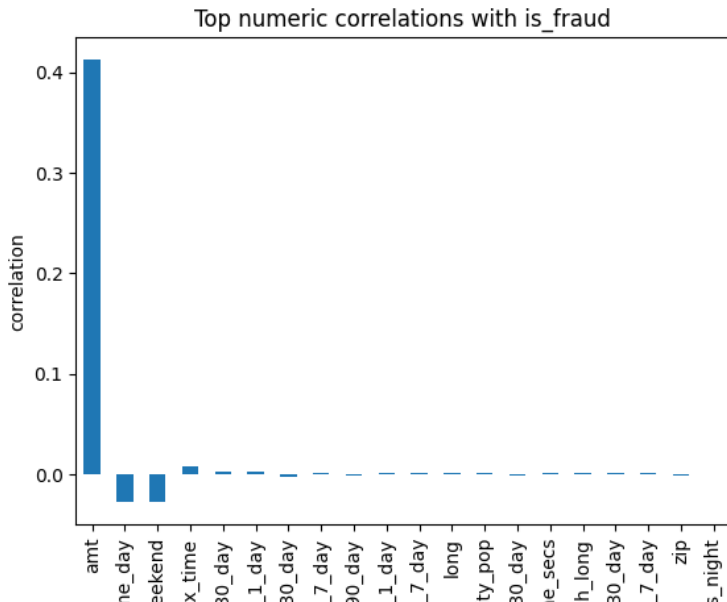


merchant_risk_30_day

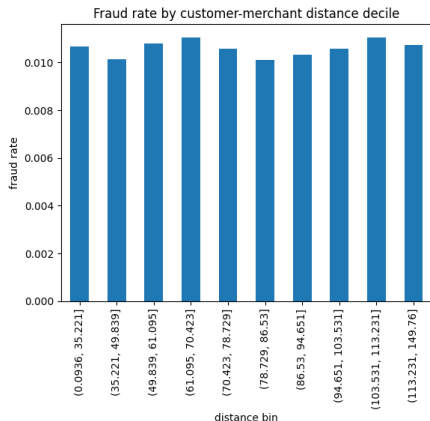


customer_num_trans_30_day

Correlation with Target



Geographic EDA: Customer–Merchant Distance



Class	Mean km	Median km
Non-fraud	76.55	78.73
Fraud	76.77	78.60

- Distance deciles are almost flat.
- Difference between fraud and non-fraud distance is negligible.
- Decision: drop distance, merchant latitude and merchant longitude.

Main findings from notebook outputs:

- Severe class imbalance: fraud rate is only 1.0597%.
- Amount is the strongest numeric signal.
- Some categories are near-deterministic in chunk 01, so cross-chunk validation is required.
- Time-of-day is weak; weekday/weekend has more visible signal.
- Job has stronger signal than merchant; merchant alone is weak.
- Customer-merchant distance is not useful in this dataset.

Modeling implication: use PR-AUC/F1, class weighting, time-based split, and cross-chunk drift checks.

Final feature set: 25 selected features used for XGBoost.

Feature groups

- Amount: amt, amount-vs-habit ratio
- Customer rolling behavior
- Merchant rolling risk
- Time indicators: day, weekend, night
- Frequency encodings for merchant and job

Dropped or controlled

- ID-like columns: card/account/profile IDs
- Raw datetime columns after feature extraction
- unix_time: only used for sorting
- High-cardinality raw text categories

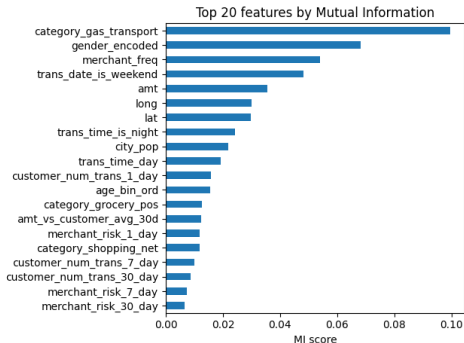
Time-based evaluation protocol per chunk:

- 64% train-inner
- 16% validation
- 20% final test

Leakage controls:

- Model never uses future rows for training.
- Threshold is selected on validation only.
- Test labels are used only for final reporting.
- `best_test_f1_diagnostic` is diagnostic only, not a deployment threshold.

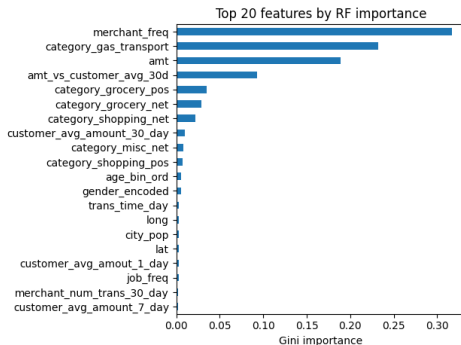
Mutual Information Ranking



Top MI signals

Feature	MI
category_gas_transport	0.0995
gender_encoded	0.0683
merchant_freq	0.0540
trans_date_is_weekend	0.0483
amt	0.0354
long	0.0300
lat	0.0297
trans_time_is_night	0.0243

RandomForest Feature Importance



Top RF importance

Feature	Importance
merchant_freq	0.3166
category_gas_transport	0.2321
amt	0.1885
amt_vs_customer_avg_30d	0.0926
category_grocery_pos	0.0349
category_grocery_net	0.0291
category_shopping_net	0.0221
customer_avg_amount_30_day	0.0100

Feature Selection Rule

Selection strategy from notebook:

- 1 Rank features by Mutual Information.
- 2 Rank features by RandomForest importance.
- 3 Keep features in top-20 of either ranking.
- 4 Remove highly collinear drop candidates.

Feature 1	Feature 2	Correlation
category_gas_transport	merchant_freq	0.9977
category_gas_transport	category_grocery_net	0.9477
customer_num_trans_1_day	velocity_num_trans_1d_30d	0.9321
category_grocery_net	merchant_freq	0.9277

Final Selected Features

Final result: 25 selected features from 41 model-ready features.

- age_bin_ord
- amt
- amt_vs_customer_avg_30d
- category_grocery_pos
- category_misc_net
- category_shopping_net
- category_shopping_pos
- city_pop
- customer_avg_amount_30_day
- customer_avg_amount_7_day
- customer_avg_amount_1_day
- customer_num_trans_30_day
- customer_num_trans_7_day
- gender_encoded
- job_freq
- lat
- long
- merchant_freq
- merchant_num_trans_30_day
- merchant_risk_1_day
- merchant_risk_30_day
- merchant_risk_7_day
- trans_date_is_weekend
- trans_time_day
- trans_time_is_night

Sanity Check: Feature Selection Impact

Feature set	Number of features	Test PR-AUC
All features	41	0.9588
Selected features	25	0.9576
Delta	-16 features	-0.0013

- Feature selection removes 39% of features with almost no PR-AUC loss.
- Smaller feature set reduces overfitting risk and speeds up XGBoost tuning.
- Selected features are used for all final XGBoost experiments.

Models

- `xgb_class_weight`
- `xgb_undersample`
- Both use tuned XGBoost parameters from Optuna.

Reason: Class weighting keeps the original data distribution and was more stable across chunks.

Imbalance handling

- Class weight: keep all data and use `scale_pos_weight`.
- Undersampling: majority class reduced to 10:1.
- Main model choice: `xgb_class_weight`.

Baseline Strategy Comparison on Chunk 01

Name	Strategy	Train size	Val PR-AUC
rf_baseline	baseline	343,209	0.9417
rf_class_weight	class weight	343,209	0.9444
rf_undersample	undersample	39,886	0.9587
xgb_baseline	baseline	343,209	0.9655
xgb_class_weight	class weight	343,209	0.9655
xgb_undersample	undersample	39,886	0.9724

- XGBoost variants outperform RandomForest variants on validation PR-AUC.
- The top two candidates selected for Optuna are xgb_undersample and xgb_class_weight.

Tuned XGBoost Parameters

Model	n_est	depth	lr	subsample	colsample	child	alpha	lambda	gamma
xgb_undersample	150	8	0.016664	0.948150	0.781684	6	0.189588	0.007976	0.660841
xgb_class_weight	400	6	0.015384	0.880095	0.754347	1	0.155126	0.005929	3.799895

Optimization objective: maximize validation PR-AUC.

Chunk 01 Final Evaluation

Model	Test PR-AUC	Threshold	F1	TN	FP	FN / TP
xgb_undersample	0.9614	0.7484	0.9564	106,356	0	75 / 823
xgb_class_weight	0.9623	0.9722	0.9564	106,356	0	75 / 823

- Both tuned XGBoost models reach precision 1.0000 and recall 0.9165 for fraud on chunk 01 test.
- xgb_class_weight has slightly higher PR-AUC and keeps the original training distribution.
- This strong chunk-01 performance motivates cross-chunk validation to check generalization.

Precision–Recall Curve on Chunk 01

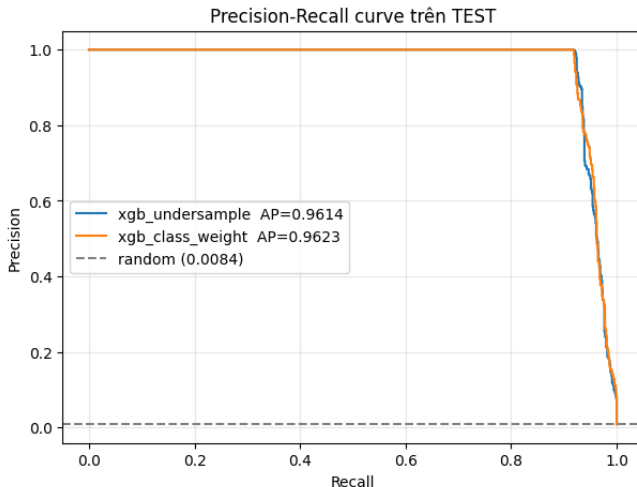


Figure: Precision–Recall curve for the two tuned XGBoost candidates on chunk 01 test set.

Why threshold matters:

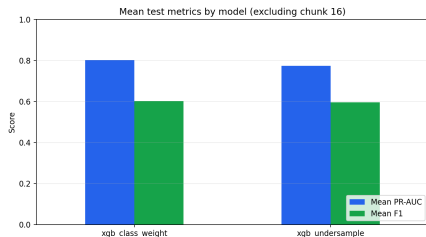
- PR-AUC measures ranking quality, independent of one fixed threshold.
- F1 depends directly on the chosen threshold.
- A low F1 can be caused by a bad threshold even if PR-AUC is high.

Dynamic thresholding:

$$\tau^* = \arg \max_{\tau} F1(y_{\text{val}}, \hat{p}_{\text{val}} \geq \tau) \quad (1)$$

The selected threshold τ^* is then applied to the final test segment.

Overall Model Comparison

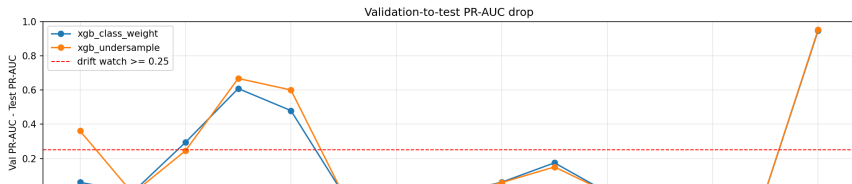
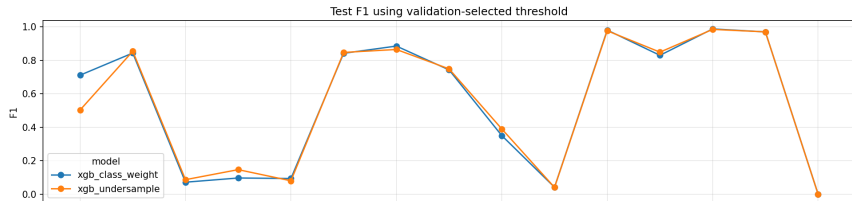
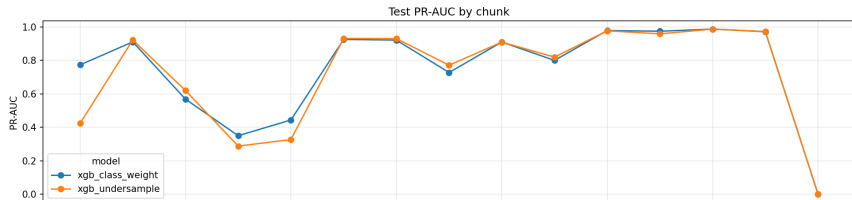


Model	PR-AUC	F1	Best
xgb_class_weight	0.8025	0.6031	0.
xgb_undersample	0.7736	0.5957	0.

Values exclude chunk 16 because the test segment has no fraud cases.

Finding: `xgb_class_weight` is the better operational baseline.

Chunk-Level Performance



Problem Chunks

Chunk	Test PR-AUC	F1 @ Val Th.	Best F1 diag.	Diagnosis
2	0.7726	0.7109	0.7365	OK
4	0.5682	0.0715	0.5996	Ranking drift suspected
5	0.3500	0.0966	0.3905	Concept/label drift likely
6	0.4439	0.0935	0.4757	Concept/label drift likely
9	0.7274	0.7427	0.7435	OK
10	0.9091	0.3500	0.8957	Threshold calibration drift
11	0.7997	0.0414	0.7559	Threshold calibration drift

Table: Diagnosis for selected low-F1 chunks using `xgb_class_weight`.

Key point: Low F1 alone is not enough to conclude concept drift.

Is It Concept Drift?

Decision rule used in this report:

- If test PR-AUC drops strongly and best diagnostic F1 remains low: **concept or label drift is likely**.
- If test PR-AUC remains high but F1 is low: **threshold/calibration drift is more likely**.
- If test set has no positive class: do not use that chunk for model-quality conclusion.

Interpretation: We separate ranking failure from threshold failure to avoid unnecessary retraining.

Concept or Label Drift Candidates

Chunks 5 and 6 are the strongest drift candidates.

- Chunk 5: validation PR-AUC = 0.9586, test PR-AUC = 0.3500.
- Chunk 6: validation PR-AUC = 0.9238, test PR-AUC = 0.4439.
- Best diagnostic F1 remains low: 0.3905 and 0.4757.
- This means the model no longer ranks fraud well in the test segment.

Action: Inspect feature distribution and label pattern, then retrain with a more recent rolling window.

Threshold Calibration Drift

Chunks 10 and 11 are mainly threshold problems.

Chunk	Test PR-AUC	F1 @ Val Th.	Best F1 diag.
10	0.9091	0.3500	0.8957
11	0.7997	0.0414	0.7559

Conclusion: The ranking is still strong, but the validation threshold is no longer suitable for the test distribution.

Issue Type Distribution

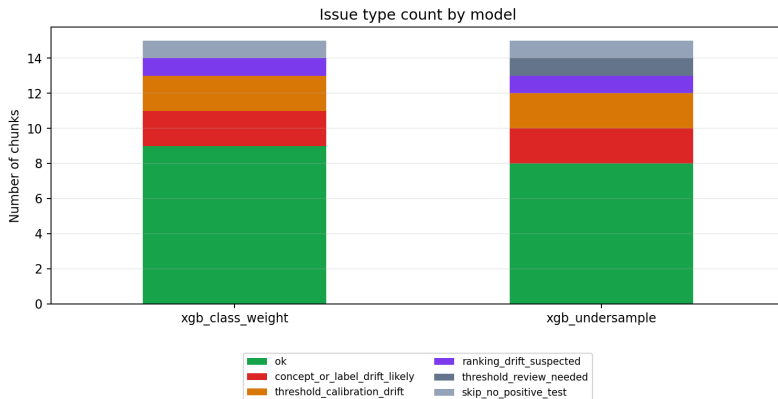


Figure: Issue categories by model across chunks 02–16.

Validation-to-Test PR-AUC Drop

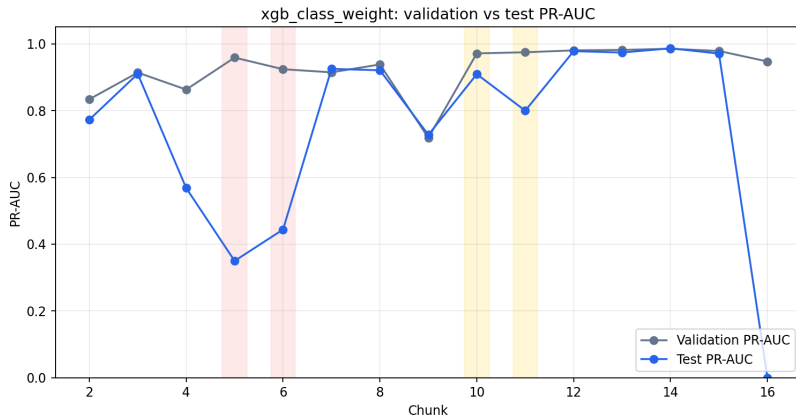


Figure: xgb_class_weight: comparison between validation and test PR-AUC.

Best practical strategy:

- 1 Use `xgb_class_weight` as the production baseline.
- 2 Use dynamic thresholding from the most recent validation window.
- 3 Use walk-forward validation instead of random split.
- 4 Retrain only when PR-AUC/ranking quality drops, not merely when F1 drops.
- 5 Monitor drift using PSI/KS on key features and fraud rate.

Reason: This separates concept drift from threshold drift and avoids unnecessary retraining.

Deployment and Monitoring Plan

Saved artifacts

- XGBoost model artifact
- Feature order metadata
- Threshold policy
- Training-window information

Monitoring

- PR-AUC and F1 by time chunk
- Fraud rate shift
- Feature PSI/KS
- Separate threshold drift vs concept drift alerts

Summary:

- XGBoost is a strong baseline for tabular fraud detection.
- `xgb_class_weight` performs better and is more stable than undersampling.
- Chunks 5–6 are likely concept/label drift candidates.
- Low F1 is not always concept drift: chunks 5–6 suggest drift, while chunks 10–11 mainly show threshold drift.
- F1 should always be interpreted together with PR-AUC and threshold diagnostics.

Final takeaway: A drift-aware pipeline should combine ranking monitoring, dynamic thresholding, and rolling retraining.

Thank You!

Questions & Discussion